

The Sentinel Shield - Near-Earth Comet (NEC) Classification

TEAM DOMINATORS

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Abstract

We present a machine learning system for automatic classification of Near-Earth Objects (NEOs) as Potentially Hazardous Objects (PHOs) using orbital and physical parameters from NASA’s NEOWS catalog (1950–2025). The model learns the physical hazard boundary defined by Minimum Orbit Intersection Distance (MOID) and Absolute Magnitude, while incorporating orbital dynamics such as eccentricity, inclination and relative velocity. A tree-based XGBoost classifier trained with class imbalance handling achieves **ROC-AUC = 0.999999** and **Recall = 0.9996**. The trained system is deployed as an interactive web application for real-time planetary defense risk assessment.

Executive Summary

What we did: Built a supervised ML model to classify Near-Earth Objects as hazardous or safe.

Performance: ROC-AUC = 0.999999, Recall = 0.9996.

Physics alignment: Model recovers NASA’s MOID–Magnitude PHO rule.

Deployment: Live web app for real-time asteroid hazard prediction.

1 Problem

Near-Earth Objects such as asteroids and comets can pose a serious threat when their orbits bring them close to Earth and their size is sufficiently large. NASA defines a Potentially Hazardous Object (PHO) as one with Minimum Orbit Intersection Distance (MOID) not exceeding 0.05 AU and Absolute Magnitude H not exceeding 22. PS-0.3 requires building a reliable and reproducible ML system that predicts this hazard status from orbital and physical parameters.

2 Dataset

We use NASA’s Near-Earth Object Web Service (NEOWS) dataset: [nasa_neows_1950_2025.csv](#). It contains 375,659 objects with physical properties, orbital elements, proximity metrics and dynamical indicators spanning 1950 to 2025.

3 Method

We remove identifiers and high-cardinality name fields, apply median imputation for missing values, log-transform skewed physical features such as diameter and velocity, and standardize all features. Because hazardous objects are statistically rare, SMOTE oversampling is used. An **XGBoost** classifier is trained to maximize recall and ROC-AUC.

4 Compute

Training was performed on an **NVIDIA RTX 3050 (6GB)** using Python, pandas, scikit-learn and XGBoost. Deployment is handled through Streamlit Cloud.

5 Results

The classifier separates hazardous and non-hazardous objects with near-perfect accuracy. A sharp probability transition is observed when MOID crosses the physical threshold of 0.05 AU, indicating that the learned decision boundary aligns with orbital mechanics.

6 Deployment

The trained model is deployed as a live interactive web application where users can test real asteroids and modify orbital parameters to observe hazard probabilities.

Live application: <https://spacecode-ps3.streamlit.app/>

7 Conclusion

We built a complete, reproducible and deployed planetary defense classifier that identifies hazardous Near-Earth Objects from real astronomical data. The Sentinel Shield system demonstrates how machine learning and orbital mechanics can be combined into a working risk-assessment interface.

Code and References

Project Repository (PS-3): [SpaceCode](#) (folder ps3/)

Dataset: [NASA NEOWS Dataset](#)

Deployment: [Sentinel Shield Live App](#)

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